

INSPECTION CERTIFICATE EN 10204-3.1
EN 10204:2005-01

DNOW L.P.
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HOUSTON TX 77041
USA

Document No.: 465-1580797/001/P
Purchase Order No.: 26769689
Benteler Order No.: 48011065
Dispatch Note No.:
Product: SEAMLESS STEEL TUBES

Inspection No.: Page: 1 / 3
Manufacturer: BSTMC Shreveport USA
Manufacturer's brand:
Stamp of the inspection representative: MI
Steelmaking process: ELECTRIC FURNACE

Terms of delivery:	ASTM-A 53-2024, ASME SA-53, ASME BPVC.II.A-2023, ASME SA-53, ASME BPVC.II.A-2025, ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2025, ASME SA-106 NDE, ASME BPVC.II.A-2025, API Specification 5L Forty-Sixth Edition April 2018 incl. Errata 1, May 2018, PSL 2, ANSI/NACE MR0175-2021/ISO 15156-1:2020, ANSI/NACE MR0175-2021/ISO 15156-2:2020, ANSI/NACE MR0103/ISO 17945:2015, ASME B36.10-2022
Dimensions-tolerances:	ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2025, ASTM-A 53-2024, ASME SA-53, ASME BPVC.II.A-2023, ASME SA-53, ASME BPVC.II.A-2025, API Specification 5L Forty-Sixth Edition April 2018 incl. Errata 1, May 2018
Steel grade:	GRADE B, GRADE C, GRADE BN, GRADE X42N
Delivery condition:	N
Product marking:	S: BENTELER - BENTELER-Sign - API 5L 1072 - API-Sign - Month/Year - A/SA 53 A/SA 106 NACE - 2,375" OD x 0,218" WT - B/C/BN/X42N - PSL2 - SMLS - Heat No. - Tube length feet - 2.970 - 15046071 - LONGHORN

AEZ = Etching ink marking FK = color marking FS = paint stenciling FSD = Color jet printer FSI = Color jet printer inside LK = Laser marking PKE = tag marking PS = die stamp TS = Ink jet spray marking

Item	Dimensions	Heat No.	Test pressure	
			PSI	sec
0020	2 NPS - SCH 80 18.000 FT - 22.500 FT	323581	2970	5

Country of melt and pour

Melted & Manufactured in USA
Gerdau, Jackson, MI, US

Heat analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0020	323581	0.160	0.260	0.95	0.004	0.002	0.12	0.02	0.06	0.15	0.004	0.001	0.001	0.0001

1. Formula: $CE_{IIW} = C + (Mn/6) + ((Cr + Mo + V)/5) + ((Cu + Ni)/15) \leq 0,43 \%$
2. Formula: $Cr + Cu + Mo + Ni + V \leq 1.00 \%$
3. Formula: $Nb + V \leq 0.06 \%$
4. Formula: $Nb + Ti + V \leq 0.15 \%$

Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0020	323581	0.361	0.354	0.005	0.006

Product analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0020	323581	0.180	0.270	0.93	0.005	0.002	0.11	0.02	0.06	0.17	0.004	0.001	0.001	0.0002
0020	323581	0.180	0.240	0.92	0.005	0.004	0.12	0.02	0.06	0.17	0.004	0.001	0.001	0.0002

1. Formula: $CE_{IIW} = C + (Mn/6) + ((Cr+Mo+V)/5) + ((Cu+Ni)/15) \leq 0.43 \%$
2. Formula: $Cr+Cu+Mo+Ni+V \leq 1.00 \%$
3. Formula: $Nb+V \leq 0.06 \%$
4. Formula: $Nb+Ti+V \leq 0.15 \%$

Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0020	323581	0.377	0.364	0.005	0.006
0020	323581	0.377	0.374	0.005	0.006

Test results

The tubes have been submitted to a leak tightness test by:		Hydrostatic test: acc. to API 5L; Test pressure/Time-record, the test records/charts shall be available for examination by the inspector	PASSED		
The tubes are non destructive tested:		ET - Test standard: acc. to API 5L; ET - Test method: acc. to ASTM-E 309; ET - Test drilled hole: 0.125 IN.; ET - Notch shape/hole: drilled hole	PASSED		
Visual inspection:	PASSED	Bending test:	PASSED	Dimensions examination:	PASSED

Results of mechanical testing

Tensile test longitudinal at RT, Strip test specimen

Item	Specimen No.	Heat No.	Specimen dimensions	Yield strength	Tensile strength	Elongation	1. Formula	Rt 0,5/Rm
Requirements			IN	RT 0,5 % PSI 42100-71800	Rm PSI 70000-95000	A2" %		MAX 0.93
0020	000011	323581	0.754 x 0.222	50328	74259	33.00	19.00	0.68
0020	003011	323581	0.752 x 0.217	52939	77450	33.00	19.00	0.68

Elongation formula

1. Formula $AF=625000 \cdot AXC^{0,2} / U^{0,9}$

Notched bar impact test [1 CHARPY_V]

Specimen position: longitudinal (L); Test temperature: -50 °F

Item	Specimen No.	Heat No.	Specimen dimensions			Absorbed energy		Impact strength	Shear fracture					
			Length	Width	Height	single	average		single	average	single	average	single	average
Requirements			mm	mm	mm	ft-lbf	ft-lbf		%	%				
			55	3,33	10,00		MIN 000							
0020	000011	323581	55	3.33	10.00	21			85					
				3.33	10.00	22			85					
				3.33	10.00	22			80	83				

Protection

outside surface varnished

Item	Heat No.	Production No.
0020	323581	15046071

Remark

NACE Standard: Hardness acc. to ANSI/NACE MR0175/ISO 15156: HRC max 22 as per ASTM A370 table 2&3., Hardness acc. to ANSI/NACE MR0103/ISO 17945: HRC max. 22 as per ASTM A370 table 2&3.; Certificate remarks: material free from mercury contamination, Material must be suitable for external fusion bond coating., Data converted from SI units, No weld repair has been carried out; Certificate-Remark: LONGHORN